

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID – 26153

Problem Statement Title – AI based Network Attack Forecasting from Network Traffic Data

Theme – Blockchain & Cybersecurity

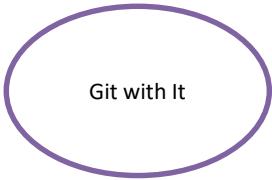
PS Category – Software

Organisation – National Technical Research Organisation (NTRO)

Team ID – [Team ID]

Team Name – Git with It





PREDICTIVE CYBER DEFENCE



We do not classify traffic — we learn how a network's state evolves, then run that model forward to see where a host is heading.

Proposed Solution

- Traffic → 60-second windows; each becomes a 39-feature snapshot of one host
- A world model learns the transition: what state usually follows this state
- Stop feeding it traffic — it generates the next 10 states on its own
- Each imagined state is scored: infiltration risk + MITRE ATT&CK stage

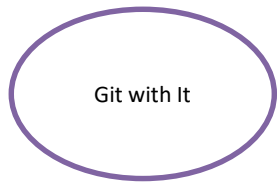
How It Addresses the Problem

- Answers “where is this host heading”, not “was that flow bad”
- 10-minute lookahead — acts before the kill chain closes
- Flow-level and packet-level features together: TTL variance, TCP window, retransmissions, scan signatures
- Every alert carries three explanations — no black box

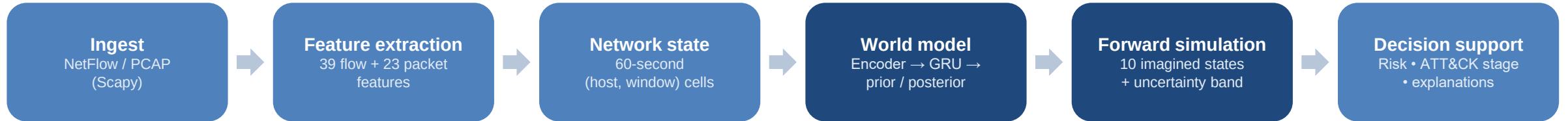
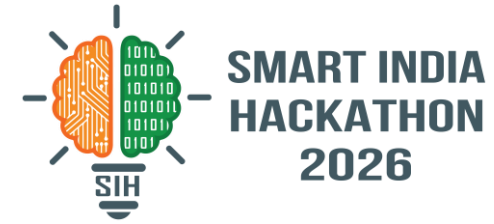
Innovation and Uniqueness

- World model, not a classifier — the heads read the latent state, so they still work on imagined states
- Provable: an imagination-only loss yields zero encoder gradient (asserted in our test suite)
- ATT&CK stages derived from CTU-13's own flow annotations, not hand-waved
- Runs fully offline on CPU — 138k parameters

Working prototype today • 258,229 network states from all 13 CTU-13 captures • 28 of 30 infected hosts flagged at 4.9% false alarms • transfers to unseen malware families • runs offline on CPU



TECHNICAL APPROACH



Technologies Used

- Python 3.13, PyTorch (CPU-only)
- Scapy — raw PCAP parsing, flow reconstruction
- pandas / NumPy — vectorised feature pipeline
- scikit-learn — baselines and metrics
- FastAPI + zero-dependency HTML dashboard
- 100% open source • no cloud, no external API

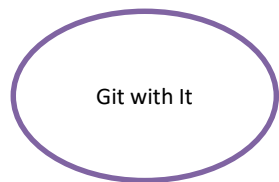
Model — Recurrent State-Space

- Encoder compresses each window into a latent state
- GRU carries context; a prior predicts the next latent without seeing traffic
- Decoder reconstructs traffic, so the latent must encode real network state
- Trained on its own rollouts, so multi-step forecasting is learned, not hoped for
- Causal attention — masked so no step sees its future

Methodology and Validation

- Dataset: CTU-13 (13 real botnet captures, CC BY)
- Split by time, never randomly — train on each capture's past, test on its future, with a guard band
- Baselines given identical features, scaler and labels
- Deterministic inference + seeded rollouts — the benchmark reproduces byte for byte
- Test suites for model shapes and PCAP header parsing

Offline dashboard: minute-by-minute replay of the forecast • kill-chain timeline • network topology • three explanation channels per prediction



FEASIBILITY AND VIABILITY



Feasibility — where the model actually wins

Stage forecast, macro-F1 by horizon	+2	+4	+10
World model (ours)	0.583	0.642	0.524
“assume nothing changes” baseline	0.473	0.474	0.436

Detection F1: 0.979 vs 0.977 — a tie

Why it is deployable

- Runs on a laptop CPU — no GPU, no cloud, no licence cost
- Consumes NetFlow/IPFIX that enterprises already export
- Sub-second triage of a full capture after first load
- Fully offline — suitable for air-gapped Critical Information Infrastructure
- MITRE stage macro-F1 0.537 vs 0.453 for the baseline
- Transfers to unseen malware families — trained on Neris and Rbot, F1 0.874 and ROC-AUC 0.982 on Virut and Murlo
- Replay reconstructs what the model knew at every minute, so an analyst can audit a decision after the fact

Potential Challenges and Risks → Our Mitigation

Label leak through metadata

The available PCAP held only botnet traffic, so a “packet data present” flag became a perfect label proxy → detected it, removed it; F1 rose 0.23 → 0.98

Detection is no better than logistic regression

0.979 vs 0.977. An earlier 0.984-vs-0.744 gap was an artefact of a test split whose positives came from one host → the model earns its place on forecasting, not detection

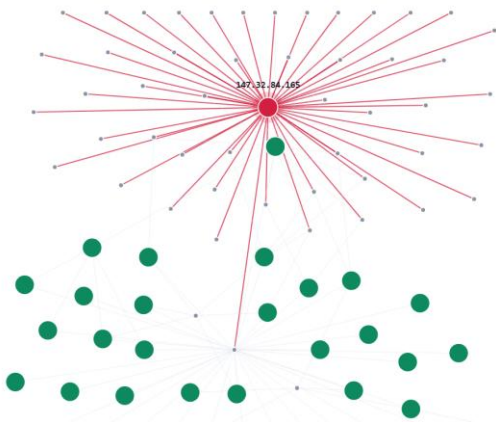
Short-burst compromise

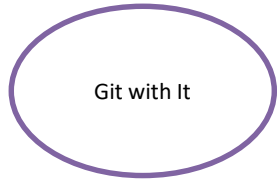
The risk head needs sustained activity → the unsupervised surprise channel covers that gap; triage flags on either, catching 28 of 30 hosts at 4.9% false alarms

Stage forecasting does not cross families

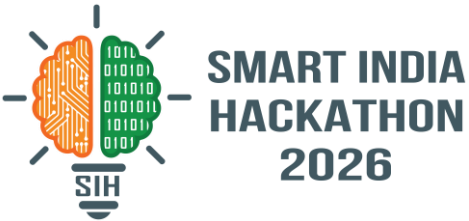
Detection transfers to unseen malware, progression does not — reported, not hidden

Live output — the compromised host and its malicious fan-out





IMPACT AND BENEFITS



Shifting the defender from reacting after compromise to acting during the kill chain

Who Benefits

- SOC analysts — a ranked queue instead of an alert flood
- Critical Information Infrastructure: power, banking, telecom, transport
- CERT-In and national response teams
- Enterprises already exporting NetFlow, with no new hardware to buy

Operational Benefits

- Up to 10 minutes of warning before the kill chain closes
- False positive rate 0.000 on the held-out period — analyst time is not wasted
- Replay lets an analyst rewind an incident minute by minute and see what the model knew, when
- Explanations make alerts auditable and challengeable
- ATT&CK stage names map onto existing playbooks

Strategic and Economic

- Fully indigenous, open-source — no foreign vendor dependency for national infrastructure
- Zero licence cost; runs on existing hardware
- Air-gap friendly — nothing leaves the network
- Earlier detection cuts breach cost, downtime and data loss

Every claim on this deck is reproducible from the repository: one command prepares the data, one trains, one benchmarks.

Datasets and Knowledge Bases

CTU-13 Dataset — Stratosphere IPS, CTU Prague

García et al., “An empirical comparison of botnet detection methods”, Computers & Security, 2014
mcfp.felk.cvut.cz/publicDatasets/CTU-13-Dataset/

MITRE ATT&CK Enterprise Matrix

Tactics TA0043, TA0001, TA0008, TA0011, TA0010
attack.mitre.org

CIC-IDS2017 / CSE-CIC-IDS2018 — UNB

Reviewed; access now gated behind registration

NCIIPC — nciipc.gov.in

Methods and Literature

World models / latent dynamics

Ha & Schmidhuber, “World Models”, NeurIPS 2018

Hafner et al., “Learning Latent Dynamics for Planning from Pixels” (PlaNet/RSSM), ICML 2019

Hafner et al., “Dream to Control” (Dreamer), ICLR 2020

Explainability

Sundararajan et al., “Axiomatic Attribution for Deep Networks” (Integrated Gradients), ICML 2017

Vaswani et al., “Attention Is All You Need”, NeurIPS 2017

Tools

Scapy • PyTorch • scikit-learn • FastAPI